

ACCREDIX — SYSTEM HIGH-LEVEL ARCHITECTURE DIAGRAM

Complete Technical Layout & File Map for NBA Criteria 3, 4, 5 & 6 Platform

1. System Architectural Component Layers

1. USER INTERFACE / PRESENTATION SHELL frontend/index.html (SPA Shell) • frontend/src/styles.css (Design Tokens, Dark Theme, Glassmorphism)		
■ (DOM Events & User Interactions)		
2. CORE CONTROLLER & REACTIVE STATE ENGINE frontend/src/app.js • Manages appState (Readiness Score: 87% → 91%), Router (switchTab), Gap Resolution & Audit Engine		
■ (Data Dispatch & View Rendering)		
3. MODULAR UI COMPONENTS • Sidebar/sidebar.js — Navigation Bar • Charts/charts.js — SVG Analytics & Trends • DashboardCard/dashboardcard.js — KPI Summary Cards • DataTable/datatable.js — Filterable Tables • FileUpload/fileupload.js — AI Doc Metadata Extractor • StatusBadge/statusbadge.js — PASS/WARNING/CRITICAL Chips	4. CRITERIA & PAGE MODULES • Dashboard/dashboard.js — Department Overview • Criterion3/criterion3.js — CO/PO Attainment & Heatmap • Criterion4/criterion4.js — Student Risk Index • Criterion5/criterion5.js — Faculty Cadre & SFR • Criterion6/criterion6.js — Publications & Evidence Chain • GapAnalysis/gapanalysis.js — Action Center Engine	5. TOOLS & UTILITIES • Evidence/evidence.js — Digital Twin Vault • VirtualInspection/virtualinspection.js — Audit Simulator • Reports/reports.js — SAR PDF Exporter • AIAssistant/aiassistant.js — Compliance Q&A; • Login/login.js — Access Control • utils/utils.js — Toast Alerts & Helpers

2. Data Lifecycle & File Execution Pipeline

Step	Stage	Primary File Target	Execution Logic
1	Page Load	frontend/index.html	Loads styles.css, UI components, page modules, and app.js.
2	State Init	frontend/src/app.js	Initializes appState object containing department metrics, student profiles, and CO/PO data.
3	User Route	src/components/Sidebar/*	Invokes switchTab(tabId) to display target view (e.g. view-criterion-3).
4	Gap Fix	src/pages/GapAnalysis/*	Clicking <i>Resolve Action</i> updates state, recalculates Readiness Score from 87% to 91%.
5	Evidence Link	src/components/FileUpload/*	Simulates AI PDF parsing, extracts metadata, links verified document to Evidence Vault.
6	Inspection	src/pages/VirtualInspection/*	Runs step-by-step compliance audit scan and computes predicted NBA grade.